로봇공학과 2025074002 임다영

Assignment 2

Problem1

VHDL Code

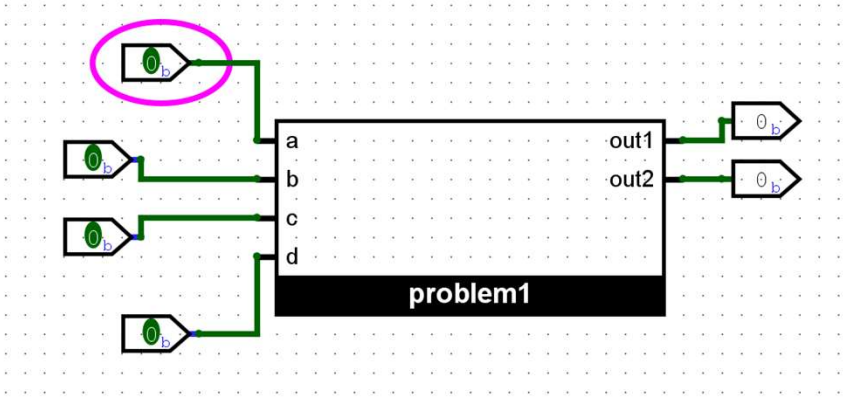
```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity problem1 is
    port (
        a, b, c, d : in std_logic;
        out1, out2 : out std_logic
    );
end entity problem1;

architecture bhv of problem1 is
begin
    out1 <= a and b and c and d;
    out2 <= a or b or c or d;
end architecture bhv;

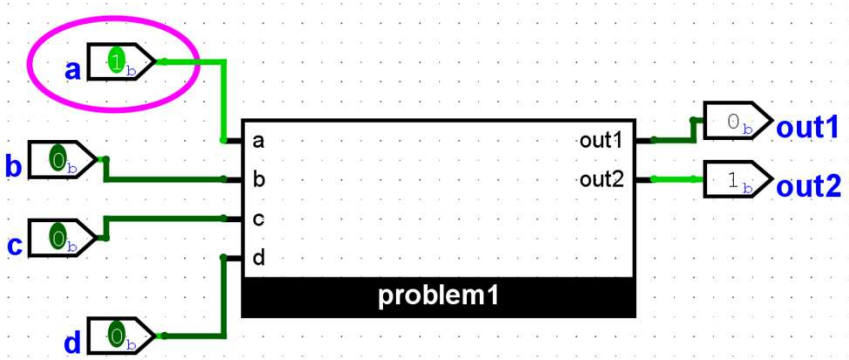
```

Circuit

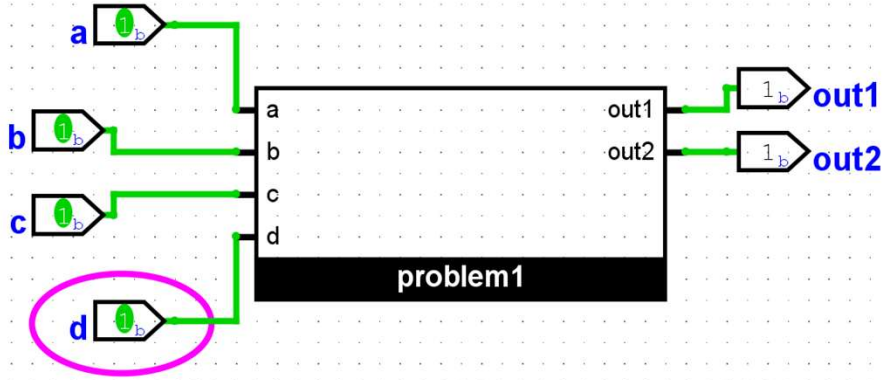


로봇공학과 2025074002 임다영

Simulation Screenshots



Simulation result for inputs (a, b, c, d) = (1, 0, 0, 0): out1 = 0, out2 = 1



Simulation result for inputs (a, b, c, d) = (1, 1, 1, 1): out1 = 1, out2 = 1

Brief Explanation of the Results

The circuit performs AND and OR operations on four inputs (a, b, c, d). When the inputs were (1, 0, 0, 0), the outputs were out1 = 0 and out2 = 1. When all inputs were 1, both outputs were 1.

These results confirm that the AND operation produces 1 only when all inputs are 1, while the OR operation produces 1 when at least one input is 1. Both simulation results matched the expected outputs.

로봇공학과 2025074002 임다영

Problem2

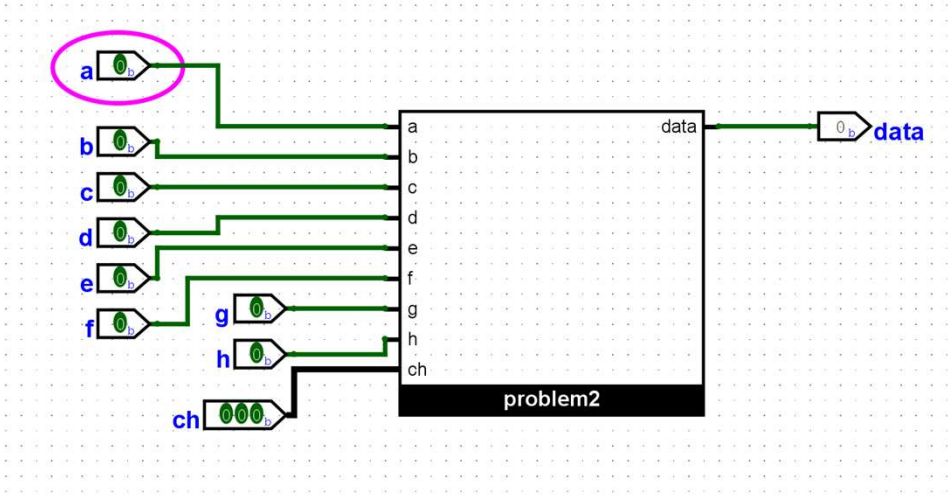
VHDL Code

```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity problem2 is
    port(
        a, b, c, d, e, f, g, h : in std_logic;
        ch : in std_logic_vector(2 downto 0);
        data : out std_logic
    );
end entity problem2;

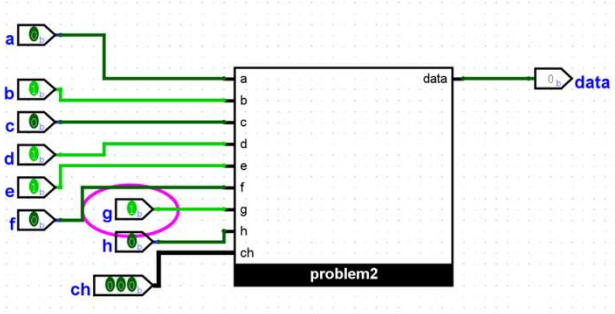
architecture bhv of problem2 is
begin
    process(a, b, c, d, e, f, g, h, ch)
    begin
        case ch is
            when "000" => data <= a;
            when "001" => data <= b;
            when "010" => data <= c;
            when "011" => data <= d;
            when "100" => data <= e;
            when "101" => data <= f;
            when "110" => data <= g;
            when "111" => data <= h;
            when others => data <= '0';
        end case;
    end process;
end architecture bhv;
```

Circuit

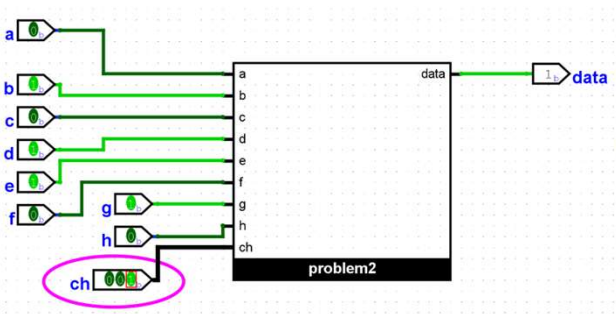


로봇공학과 2025074002 임다영

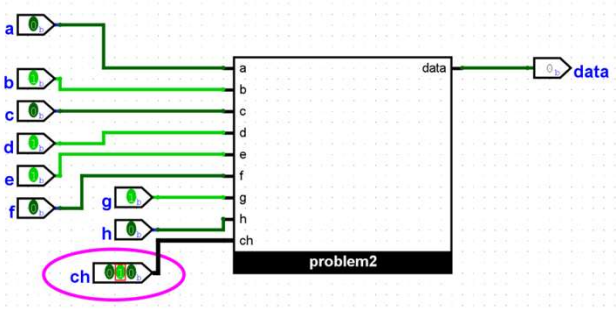
Simulation Screenshots



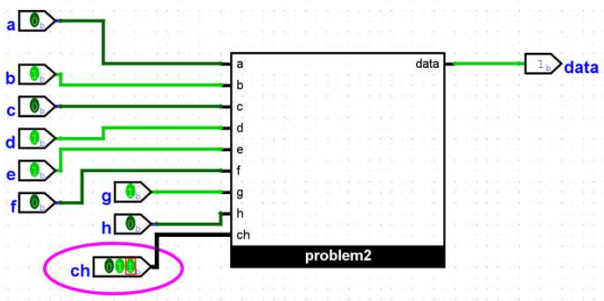
ch = 000, selected input a = 0, output data = 0



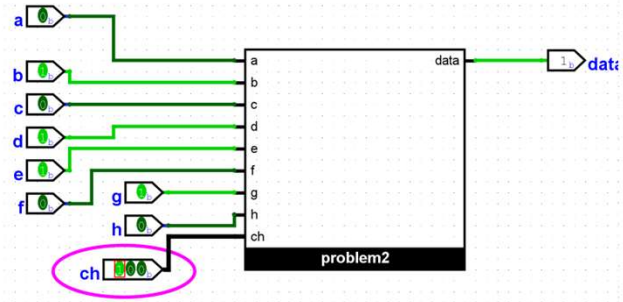
ch = 001, selected input b = 1, output data = 1



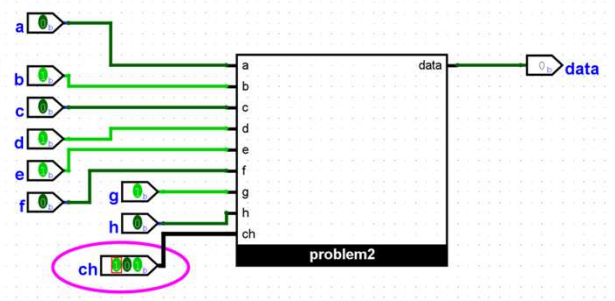
ch = 010, selected input c = 0, output data = 0



ch = 011, selected input d = 1, output data = 1



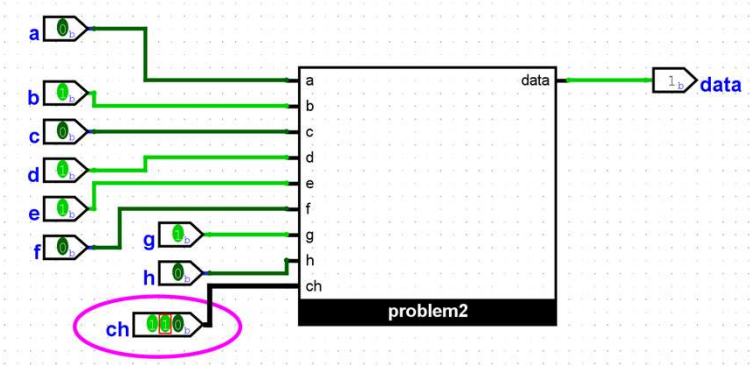
ch = 100, selected input e = 1, output data = 1



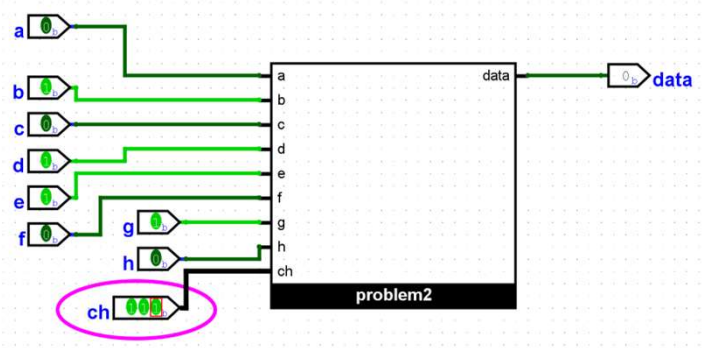
ch = 101, selected input f = 0, output data = 0

로봇공학과 2025074002 임다영

Simulation Screenshots



ch = 110, selected input g = 1, output data = 1



ch = 111, selected input h = 0, output data = 0

Brief Explanation of the Results

The circuit implements an 8-to-1 multiplexer that selects one of eight data inputs (a–h) according to a 3-bit selection signal (ch).

During the simulation, the selection signal was changed from 000 to 111 to verify the output for each input. When ch = 000, input a was selected and the output was 0. When ch = 001, input b was selected and the output was 1. The remaining cases also produced the values of the corresponding selected inputs.

These results confirm that the multiplexer correctly selects one of the eight inputs according to the 3-bit selection signal, and the simulation results match the expected behavior defined in the VHDL code.

로봇공학과 2025074002 임다영

Problem3

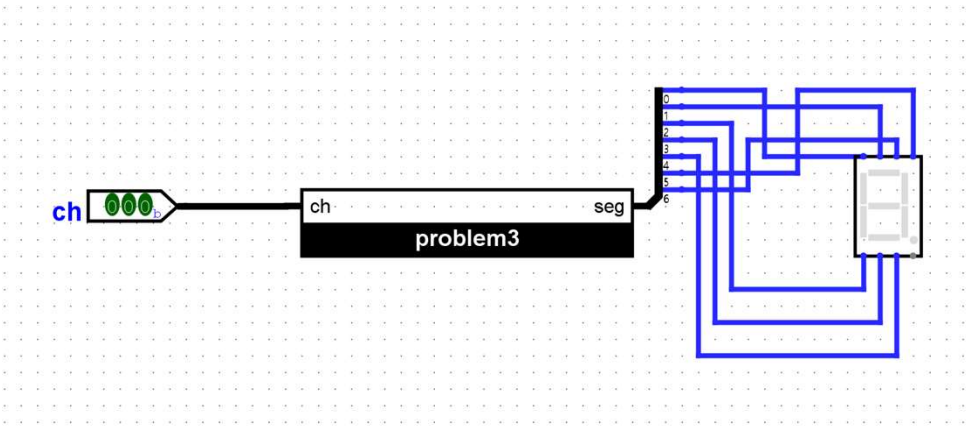
VHDL Code

```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity problem3 is
    port(
        ch  : in std_logic_vector(2 downto 0);
        seg : out std_logic_vector(6 downto 0)
    );
end entity problem3;

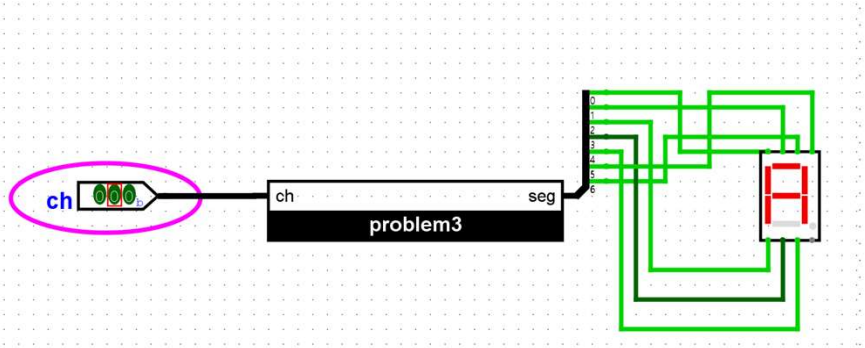
architecture bhv of problem3 is
begin
    process(ch)
    begin
        case ch is
            when "000" => seg <= "1110111"; -- a
            when "001" => seg <= "0011111"; -- b
            when "010" => seg <= "0001101"; -- c
            when "011" => seg <= "0111101"; -- d
            when "100" => seg <= "1000111"; -- f
            when "101" => seg <= "1111011"; -- g
            when others => seg <= "0000000";
        end case;
    end process;
end architecture bhv;
```

Circuit

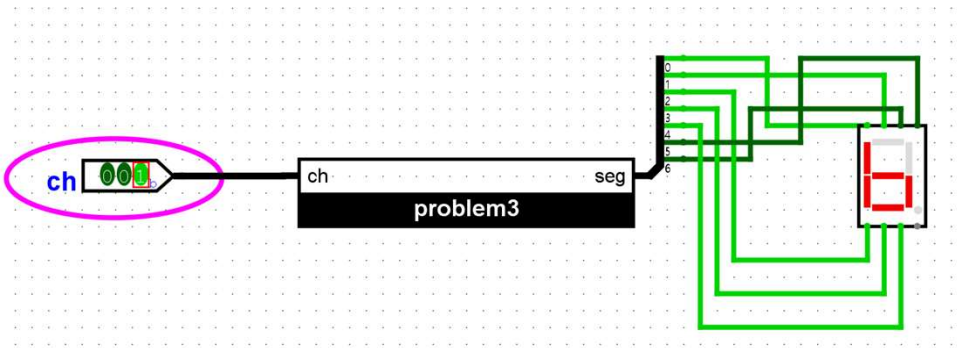


로봇공학과 2025074002 임다영

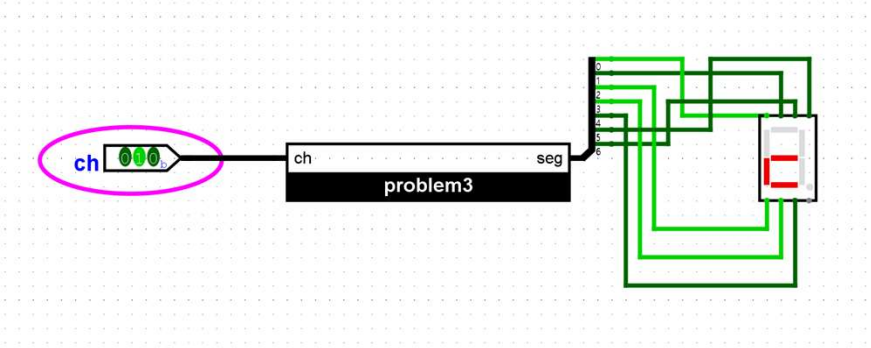
Simulation Screenshots



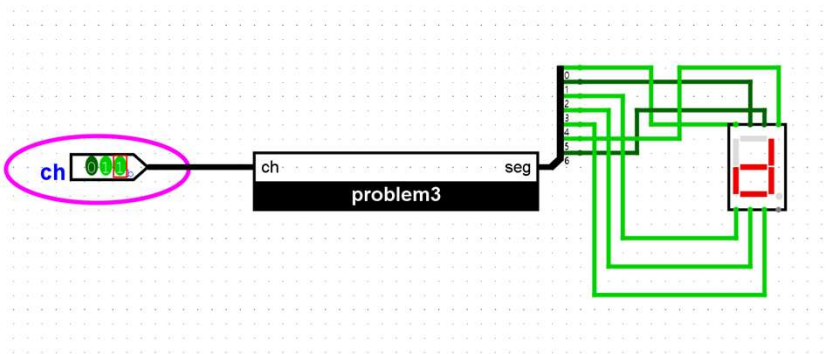
ch = 000, 7-segment display output: a



ch = 001, 7-segment display output: b



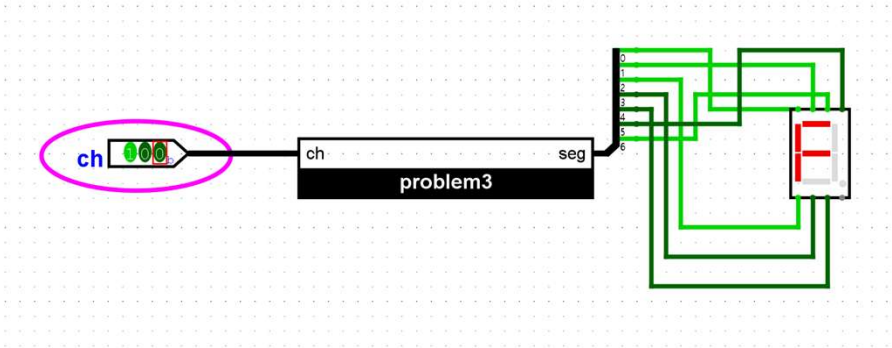
ch = 010, 7-segment display output: c



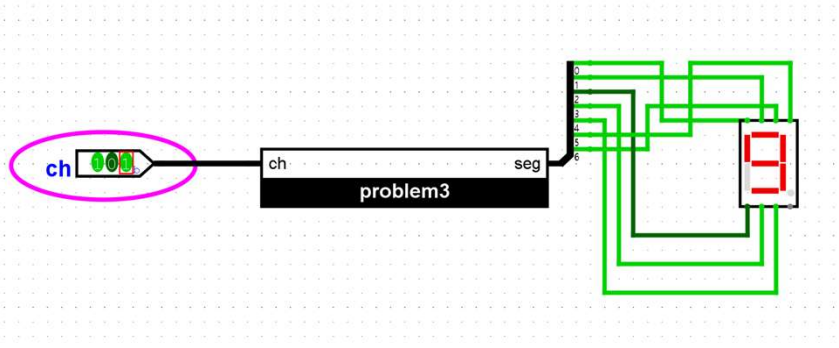
ch = 011, 7-segment display output: d

로봇공학과 2025074002 임다영

Simulation Screenshots



ch = 100, 7-segment display output: f



ch = 101, 7-segment display output: g

Brief Explanation of the Results

The circuit was designed to display six letters (a, b, c, d, f, g) on a 7-segment display according to a 3-bit selection signal (ch).

During the simulation, the selection signal was changed from 000 to 101 to verify the corresponding display outputs. When ch = 000, the letter a was displayed, and when ch = 001, the letter b was displayed. The remaining selection signals correctly displayed c, d, f, and g, respectively.

These results confirm that the VHDL module generates the appropriate 7-bit output patterns according to the selection signal and successfully displays the corresponding letters on the 7-segment display.

로봇공학과 2025074002 임다영

Problem4

VHDL Code

```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity problem4 is
    port(
        vt     : in std_logic_vector(24 downto 0);
        result : out std_logic
    );
end entity problem4;

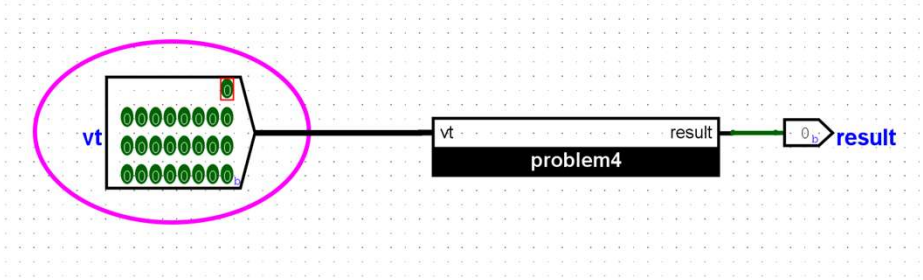
architecture bhv of problem4 is
begin
    process(vt)
        variable count : integer range 0 to 25;
    begin
        count := 0;

        for i in 0 to 24 loop
            if vt(i) = '1' then
                count := count + 1;
            end if;
        end loop;

        if count >= 13 then
            result <= '1';
        else
            result <= '0';
        end if;
    end process;
end architecture bhv;

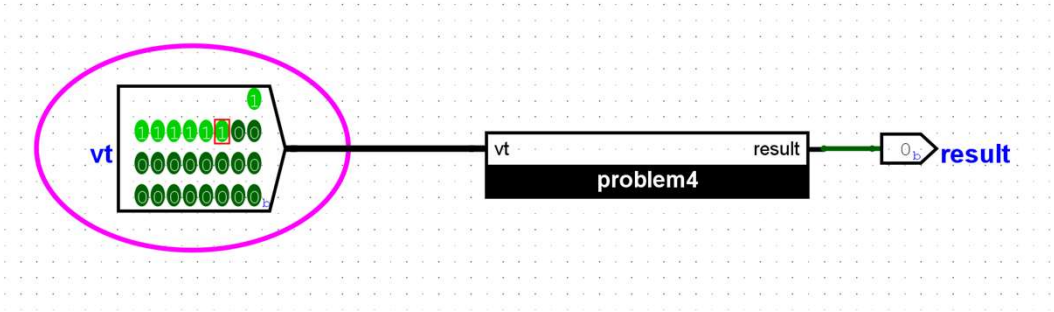
```

Circuit

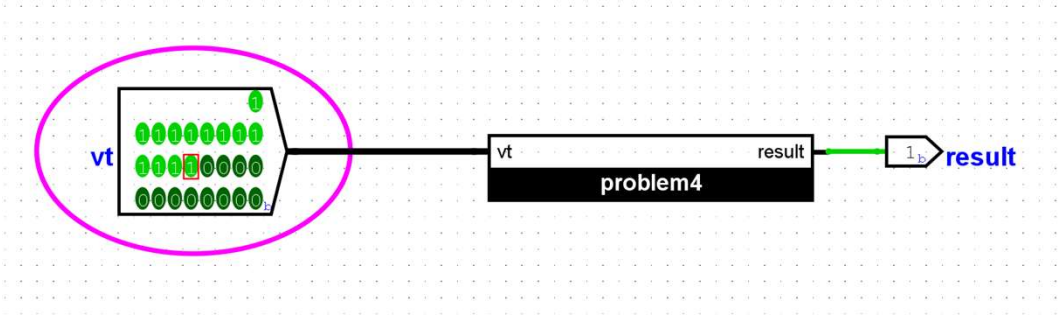


로봇공학과 2025074002 임다영

Simulation Screenshots



Simulation result with 12 votes in favor: result = 0



Simulation result with 13 votes in favor: result = 1

Brief Explanation of the Results

The circuit implements a 25-person voting machine that determines whether a majority of voters are in favor. Each vote is represented by a single bit, where 1 indicates approval and 0 indicates disapproval.

During the simulation, two boundary cases were tested to verify the majority decision. When 12 voters were in favor, the output was result = 0 because the majority condition was not satisfied. When 13 voters were in favor, the output changed to result = 1, indicating that a majority had been reached.

These results confirm that the circuit correctly counts the number of 1s in the 25-bit input and outputs 1 only when at least 13 voters are in favor. The simulation results match the majority decision condition defined in the VHDL code.

로봇공학과 2025074002 임다영

Problem5

VHDL Code

```vhdl
library ieee;
use ieee.std_logic_1164.all;

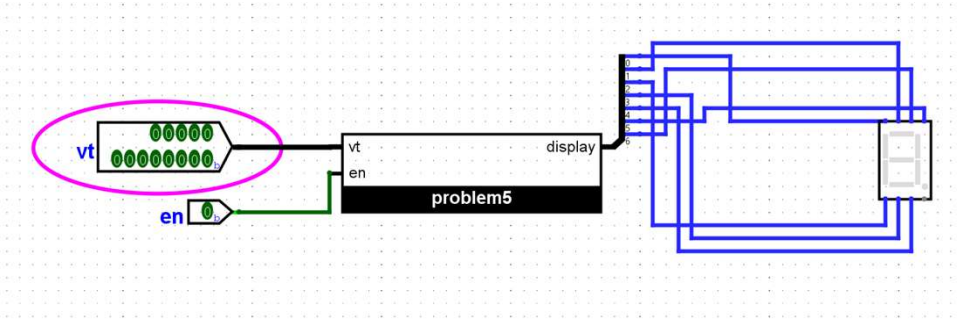
entity problem5 is
    port(
        vt      : in std_logic_vector(12 downto 0);
        en      : in std_logic;
        display : out std_logic_vector(6 downto 0)
    );
end entity problem5;

architecture bhv of problem5 is
begin
    process(vt, en)
        variable count : integer range 0 to 13;
    begin
        count := 0;

        for i in 0 to 12 loop
            if vt(i) = '1' then
                count := count + 1;
            end if;
        end loop;

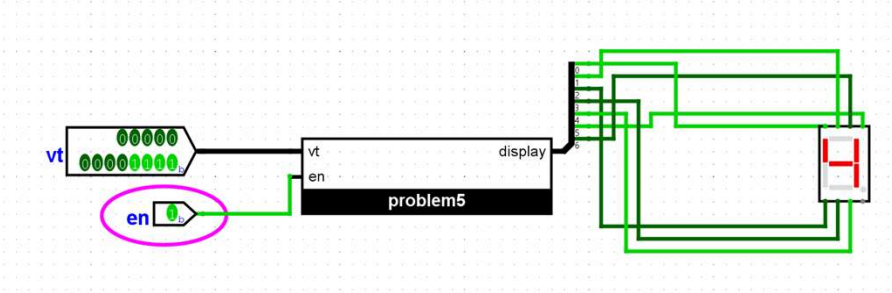
        if en = '0' then
            display <= "0000000";
        else
            case count is
                when 0  => display <= "1111110"; -- 0
                when 1  => display <= "0110000"; -- 1
                when 2  => display <= "1101101"; -- 2
                when 3  => display <= "1111001"; -- 3
                when 4  => display <= "0110011"; -- 4
                when 5  => display <= "1011011"; -- 5
                when 6  => display <= "1011111"; -- 6
                when 7  => display <= "1110000"; -- 7
                when 8  => display <= "1111111"; -- 8
                when 9  => display <= "1111011"; -- 9
                when 10 => display <= "1110111"; -- A
                when 11 => display <= "0011111"; -- b
                when 12 => display <= "1001110"; -- C
                when 13 => display <= "0111101"; -- d
                when others => display <= "0000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

Circuit

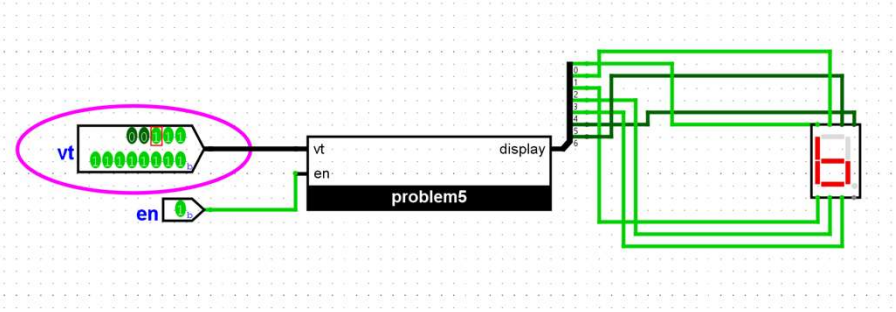


로봇공학과 2025074002 임다영

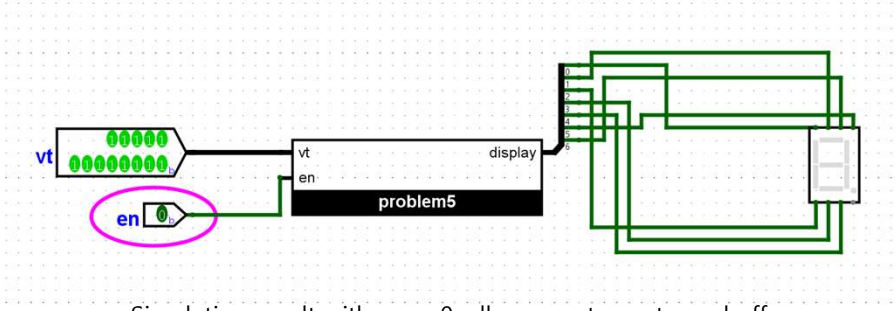
Simulation Screenshots



Simulation result with 4 active bits and en = 1: the digit 4 is displayed



Simulation result with 11 active bits and en = 1: the letter b is displayed



Simulation result with en = 0: all segments are turned off

Brief Explanation of the Results

The circuit was designed to count the number of 1s in a 13-bit input signal (vt) and display the corresponding digit or letter on a 7-segment display. An enable signal (en) was also implemented to control whether the display is turned on or off.

During the simulation, when en = 1 and the input contained four active bits, the digit 4 was displayed. When the input contained eleven active bits, the letter b was displayed. Additionally, setting en = 0 turned off all segments regardless of the input value.

These results confirm that the circuit displays the corresponding digit or letter based on the number of 1s in the input signal and correctly controls the display using the enable signal. The simulation results match the expected behavior defined in the VHDL code.